



50 Advanced MCQs

Python Core & Advanced

1. What will be the output of the following?

```
def func(a, L=[]):  
    L.append(a)  
    return L
```

```
print(func(1))  
print(func(2))
```

- A. [1], [2]
- B. [1], [1,2]
- C. [1], [2,1]
- D. Error

Answer: B — [1], [1,2]

2. What is the average time complexity of searching in a Python dictionary?

- A. $O(n)$
- B. $O(\log n)$
- C. $O(1)$
- D. $O(n \log n)$

Answer: C — $O(1)$

3. Which statement best describes Python decorators?

- A. Modify variables globally
- B. Wrap functions to extend behavior
- C. Replace functions permanently
- D. Only used in classes

Answer: B — Wrap functions to extend behavior

4. What does the expression `lambda x: x**2` produce?

- A. Integer
- B. Function object
- C. List
- D. Generator

Answer: B — Function object

5. Which statement about recursion in Python is correct?

- A. It does not use stack memory
- B. It is always faster than loops
- C. It uses call stack memory
- D. It cannot return values

Answer: C — It uses call stack memory

OOP Concepts

6. Which principle is primarily responsible for data hiding?

- A. Abstraction
- B. Encapsulation
- C. Inheritance
- D. Polymorphism

Answer: B — Encapsulation

7. Method overriding occurs when:

- A. Same method exists in same class
- B. Same method exists in parent and child class
- C. Different method names are used
- D. Static methods are used

Answer: B — Same method exists in parent and child class

8. What does MRO stand for in Python?

- A. Memory Runtime Object
- B. Method Resolution Order
- C. Module Runtime Operation
- D. Multi Reference Object

Answer: B — Method Resolution Order

Multithreading & Performance

9. Why is multithreading limited in Python?

- A. Syntax limitations
- B. Memory issues
- C. Global Interpreter Lock
- D. CPU architecture

Answer: C — Global Interpreter Lock

10. Multiprocessing is most suitable for:

- A. I/O-bound tasks
- B. CPU-bound tasks
- C. UI rendering
- D. Logging

Answer: B — CPU-bound tasks

NumPy & Pandas

11. What does `.iloc` in Pandas represent?

- A. Label-based indexing
- B. Integer-based indexing
- C. Boolean indexing
- D. Mixed indexing

Answer: B — Integer-based indexing

12. Which operation demonstrates vectorization?

- A. Loop-based addition
- B. NumPy array addition
- C. File reading
- D. String parsing

Answer: B — NumPy array addition

13. What is NaN in data science?

- A. String
- B. Missing numeric value
- C. Boolean
- D. Integer

Answer: B — Missing numeric value

APIs & Backend

14. Why is FastAPI widely used?

- A. Only synchronous support
- B. No validation
- C. Async support and performance
- D. Requires Java

Answer: C — Async support and performance

15. What does HTTP status code 404 indicate?

- A. Success
- B. Unauthorized
- C. Not Found
- D. Server Error

Answer: C — Not Found

16. Which HTTP method is idempotent?

- A. POST
- B. PUT
- C. PATCH
- D. CONNECT

Answer: B — PUT

17. REST stands for:

- A. Random Execution System Transfer
- B. Representational State Transfer
- C. Resource Execution Style Transfer
- D. Runtime Execution System Transfer

Answer: B — Representational State Transfer

Databases

18. Which is not a SQL operation?

- A. SELECT
- B. INSERT
- C. MAPREDUCE
- D. DELETE

Answer: C — MAPREDUCE

19. MongoDB stores data as:

- A. Tables
- B. Rows
- C. Documents
- D. Graphs

Answer: C — Documents

20. ORM is used for:

- A. UI design
- B. Database abstraction
- C. Networking
- D. Logging

Answer: B — Database abstraction

DevOps

21. Docker is primarily used for:

- A. Coding
- B. Containerization
- C. Testing
- D. Debugging

Answer: B — Containerization

22. CI/CD stands for:

- A. Continuous Integration / Continuous Deployment
- B. Code Integration / Data Deployment
- C. Continuous Inspection
- D. None

Answer: A — Continuous Integration / Continuous Deployment

23. Which Git command stages changes?

- A. git push
- B. git commit
- C. git add
- D. git clone

Answer: C — git add

Statistics

24. Mean is sensitive to:

- A. Median
- B. Outliers
- C. Mode
- D. Frequency

Answer: B — Outliers

25. Standard deviation measures:

- A. Accuracy
- B. Spread of data
- C. Mean
- D. Probability

Answer: B — Spread of data

26. Probability values lie between:

- A. -1 to 1
- B. 0 to 1
- C. 0 to 100
- D. $-\infty$ to ∞

Answer: B — 0 to 1

Machine Learning

27. Linear regression is used for:

- A. Classification
- B. Regression
- C. Clustering
- D. Reinforcement

Answer: B — Regression

28. Logistic regression is mainly used for:

- A. Regression
- B. Classification
- C. Clustering
- D. Dimensionality reduction

Answer: B — Classification

29. Overfitting refers to:

- A. Under learning
- B. Model learning noise
- C. Simple model
- D. No training

Answer: B — Model learning noise

30. Train-test split helps in:

- A. Increasing data
- B. Preventing biased evaluation
- C. Reducing features
- D. Deployment

Answer: B — Preventing biased evaluation

Clustering

31. KMeans requires:

- A. Labels
- B. Number of clusters
- C. Supervision
- D. Trees

Answer: B — Number of clusters

32. DBSCAN is useful for:

- A. Linear data
- B. Handling noise
- C. Regression
- D. Time series

Answer: B — Handling noise

Evaluation Metrics

33. Accuracy is unreliable in:

- A. Balanced dataset
- B. Imbalanced dataset
- C. Small dataset
- D. Large dataset

Answer: B — Imbalanced dataset

34. Precision focuses on:

- A. All predictions
- B. Correct positive predictions
- C. Negative predictions
- D. Recall

Answer: B — Correct positive predictions

Pipelines

35. Machine learning pipelines help in:

- A. Visualization
- B. Automating workflow

- C. Data deletion
- D. Model storage

Answer: B — Automating workflow

Deep Learning

36. Activation functions introduce:

- A. Linearity
- B. Non-linearity
- C. Errors
- D. Data

Answer: B — Non-linearity

37. ReLU stands for:

- A. Random Linear Unit
- B. Rectified Linear Unit
- C. Reduced Learning Unit
- D. None

Answer: B — Rectified Linear Unit

38. CNN is best suited for:

- A. Text
- B. Images
- C. Audio
- D. Numbers

Answer: B — Images

39. Overfitting in deep learning can be reduced using:

- A. Dropout
- B. More layers
- C. Less data
- D. Higher learning rate

Answer: A — Dropout

Advanced Concepts

40. Feature scaling means:

- A. Data removal
- B. Normalizing data
- C. Encoding labels
- D. Cleaning data

Answer: B — Normalizing data

41. Gradient descent is used for:

- A. Data loading
- B. Optimization
- C. Visualization
- D. Storage

Answer: B — Optimization

42. Learning rate controls:

- A. Model size
- B. Step size
- C. Dataset size
- D. Accuracy

Answer: B — Step size

Scenario-Based

43. Which model is suitable for email spam detection?

- A. Linear regression
- B. Logistic regression
- C. KMeans
- D. PCA

Answer: B — Logistic regression

44. Best algorithm for customer segmentation:

- A. Logistic regression
- B. KMeans

- C. Linear regression
- D. CNN

Answer: B — KMeans

45. High training accuracy but low test accuracy indicates:

- A. Underfitting
- B. Overfitting
- C. Balanced model
- D. Optimized model

Answer: B — Overfitting

System Design

46. Which tool is used for API testing?

- A. Docker
- B. Postman
- C. Git
- D. NumPy

Answer: B — Postman

47. JSON is:

- A. Database
- B. Data format
- C. Programming language
- D. Protocol

Answer: B — Data format

48. FastAPI uses which library for validation?

- A. Flask
- B. Pydantic
- C. Django
- D. SQL

Answer: B — Pydantic

Practical ML

49. Which technique improves generalization?

- A. Noise addition
- B. Regularization
- C. Overfitting
- D. Random weights

Answer: B — Regularization

50. Best way to deploy ML models in production:

- A. Local script
- B. REST API
- C. Excel
- D. Text file

Answer: B — REST API